



Hybrid Classical–Quantum Split Federated Learning for Intrusion Resilience (HCQ-SFL-IR)

Dr. Sumaiya Tabassum Nimi

Assistant Professor, Dept. of ECE — North South University

Application Vision

Modern smart healthcare ecosystems—including hospitals, telemedicine platforms, wearable devices, and remote patient monitoring systems—are increasingly dependent on distributed artificial intelligence models for diagnosis, monitoring, and clinical decision support. These systems, however, are highly vulnerable to cyber threats such as compromised Internet-of-Medical-Things (IoMT) devices, poisoned model updates, insider attacks, adversarial manipulation of clinical signals, and regulatory violations caused by unintended data leakage.

The proposed **Quantum-Resilient Security-as-a-Service (QR-SecaaS)** layer is a plug-and-play security middleware built on top of the Hybrid Classical–Quantum Split Federated Learning for Intrusion Resilience (HCQ-SFL-IR) framework. It continuously protects collaborative healthcare AI pipelines without accessing raw patient data, thereby preserving privacy and regulatory compliance.

Security-as-a-Service Paradigm

Unlike traditional cybersecurity mechanisms such as firewalls, intrusion detection systems, or encryption layers, QR-SecaaS secures the *learning process itself*. The service operates at the level of model updates and feature activations rather than network packets. It is deployed as a cloud- or hospital-hosted service and requires no modification to patient-facing devices or clinical workflows.

Healthcare providers subscribe to QR-SecaaS as a managed service that transparently secures federated and split learning pipelines while remaining compliant with healthcare data regulations such as HIPAA and GDPR.

Deployment Architecture

Edge and Device Layer

This layer includes wearable sensors, bedside monitors, ICU systems, and medical imaging edge servers. Each device runs a lightweight classical feature extractor, such as a compact CNN, MLP, or Tiny Transformer. Raw patient data remains local to the device or institution, and only intermediate feature activations are transmitted.

This design ensures that patient data never leaves institutional boundaries, and even compromised devices cannot leak sensitive clinical information.

QR-SecaaS Cloud Layer

The cloud layer constitutes the core Security-as-a-Service offering. It provides the following modules:

- **Quantum Feature Scrambling Engine:** Encodes classical activations into quantum states, preventing data reconstruction and inversion.

- **Quantum Intrusion Detection Module:** Detects poisoned or anomalous updates using quantum fidelity and trace distance metrics.
- **Trust-Aware Federated Aggregator:** Assigns trust scores to participants and suppresses malicious contributions.
- **Self-Healing Controller:** Dynamically adapts security strictness, quarantines suspicious participants, and increases quantum embedding depth during attacks.

Healthcare AI Applications

Clinical applications such as disease risk prediction, early warning systems for sepsis or arrhythmia, remote patient monitoring dashboards, and clinical decision support tools consume globally trained models secured by QR-SecaaS. These applications remain agnostic to the underlying quantum security mechanisms.

Representative Use-Case Scenarios

Multi-Hospital Collaborative Model Training

Multiple hospitals collaboratively train a cardiac risk prediction model. If one hospital becomes compromised and injects poisoned updates, QR-SecaaS detects abnormal quantum fidelity scores, down-weights or rejects the malicious contribution, and quarantines the hospital. The global model remains accurate and uncompromised without manual intervention.

Wearable Device Tampering

In the event of adversarial manipulation of wearable ECG signals, QR-SecaaS detects anomalous quantum embeddings, isolates the compromised device, decays its trust score, and alerts clinicians. The attack is contained at the device level, ensuring patient safety.

Insider Threat Detection

When an insider attempts to manipulate model updates to bias clinical outcomes, QR-SecaaS identifies subtle deviations from quantum reference states, restricts access privileges, and prevents corruption of the global model—despite the absence of known attack signatures.

Service Model and API Design

QR-SecaaS is offered as a managed service hosted by trusted entities such as national health authorities, government-backed medical research networks, or certified cloud providers.

The service exposes secure APIs, including:

- `/secure-activate`
- `/quantum-score`
- `/federated-aggregate`
- `/intrusion-alert`

- /trust-report

Healthcare systems can integrate these APIs without modifying their existing AI pipelines.

Regulatory and Ethical Compliance

QR-SecaaS supports compliance with HIPAA, GDPR, and emerging medical AI governance frameworks. No raw patient data is transmitted or stored externally, and quantum embeddings ensure non-reconstructibility by design. Trust reports and intrusion logs are auditable, supporting transparency and accountability.

Advantages Over Existing Healthcare Security Solutions

Feature	Traditional IDS	Secure FL	QR-SecaaS
Protects model training	No	Partial	Yes
Zero-day resilience	No	No	Yes
Raw data sharing	No	No	No
Insider attack detection	No	No	Yes
Self-healing capability	No	No	Yes
Quantum-enhanced privacy	No	No	Yes

Societal and Clinical Impact

QR-SecaaS prevents AI-driven misdiagnosis caused by cyber attacks, enables nation-scale collaborative healthcare AI, increases trust in clinical decision support systems, and prepares healthcare infrastructures for post-quantum cyber threats while preserving patient privacy.

Research and Commercial Outcomes

From a research perspective, QR-SecaaS enables multiple high-impact journal publications and establishes a benchmark for quantum-secure healthcare AI. From a translational perspective, it offers a hospital-ready security layer with strong potential for commercialization, national deployment, and spin-off ventures.

Positioning Statement

QR-SecaaS is a quantum-resilient, privacy-preserving Security-as-a-Service layer that protects smart healthcare AI systems against poisoning, insider threats, and zero-day cyber intrusions—without accessing patient data.

Dataset Strategy for HCQ-SFL-IR: Medical IoMT and Intrusion-Resilient Evaluation

To rigorously evaluate the proposed *Hybrid Classical-Quantum Split Federated Learning for Intrusion Resilience (HCQ-SFL-IR)* framework, we identify a suite of medical IoMT datasets and network

intrusion datasets that collectively satisfy the following criteria: (i) real-world sensor heterogeneity, (ii) multi-institution or multi-device partitions, (iii) strong non-IID characteristics, (iv) suitability for both health prediction and adversarial intrusion experiments, and (v) alignment with the federated split learning paradigm.

Primary Medical IoMT Dataset

MIMIC-III / MIMIC-IV (Waveform, Vitals, Laboratory Data)

MIMIC-III and MIMIC-IV provide comprehensive, high-resolution ICU signals including ECG, PPG, blood pressure, respiration, and a variety of vital signs and laboratory measurements. These datasets are ideal for the HCQ-SFL-IR architecture due to:

- **Realistic IoMT environment:** Multi-sensor ICU bedside monitoring with natural device heterogeneity.
- **Federated partitioning flexibility:**
 - across hospitals,
 - across ICU units,
 - across individual ICU beds, or
 - across sensor types (e.g., ECG node, BP node, SpO₂ node).
- **Intrinsic suitability for intrusion-resilience testing:**
 - corrupted or manipulated vital-sign segments,
 - poisoned temporal trends,
 - malicious outlier injections,
 - sensor spoofing via synthetic ECG/PPG signals.

Fit to HCQ-SFL-IR: Client-side lightweight CNN/TCN encoders process raw signals, server-side PQC's evaluate quantum fidelity and trace-distance anomalies, and the federated aggregator incorporates quantum-derived trust scores. The dataset provides a realistic foundation for demonstrating clinical applicability.

Secondary IoMT Dataset (Wearable Health Monitoring)

WESAD

WESAD contains multimodal wearable sensor data, including EDA, EMG, temperature, respiration, and accelerometer signals. It is well-suited for:

- simulating on-body IoMT devices,
- stress and affect detection tasks,
- multi-sensor federated configurations,
- adversarial evaluations such as abnormal-segment injection and drift-based poisoning.

Tertiary IoMT Dataset (Activity Monitoring)

UCI HAR / PAMAP2

UCI HAR and PAMAP2 provide motion-sensor data collected from smartphones and wearable IMUs. These datasets support:

- baseline evaluations of edge-level federated learning systems,
- assessment of generalizability across device categories,
- scalability experiments under lightweight settings.

Network-Level Intrusion Detection Datasets

Since intrusion resilience is a central objective of HCQ-SFL-IR, we incorporate realistic IoT network attack datasets widely used in cybersecurity-focused FL literature.

TON_IoT (Healthcare Subset)

TON_IoT includes IoMT traffic traces with numerous attack types, such as botnets, DDoS, privilege escalation, and sensor spoofing. It supports:

- federated intrusion detection experiments,
- quantum anomaly scoring across nodes,
- multi-client simulations where nodes represent hospital wards or device types.

IoTID20

Offers real IoT network traffic with diverse attack labels. Useful for validating the robustness of PQC-based anomaly detection.

CIC-IDS2017

A widely used benchmark containing replay attacks, scan attempts, brute-force attempts, infiltration, and data exfiltration scenarios. Suitable for benchmarking HCQ-SFL-IR against classical IDS baselines.

Recommended Dataset Combination

To maximize clinical relevance, security coverage, and methodological novelty, we propose the following dataset suite:

- **Primary medical IoMT:** MIMIC-III Waveform / Vitals
- **Secondary wearable IoMT:** WESAD
- **Network intrusion:** TON_IoT Healthcare Subset
- **Optional additions:** IoTID20, CIC-IDS2017

This combination demonstrates that HCQ-SFL-IR operates effectively across diverse IoMT ecosystems, integrates quantum-classical split learning seamlessly, and maintains robustness at both the physiological-signal and network-traffic levels.

Attack Scenarios

To fully assess adversarial robustness, we incorporate a comprehensive suite of sensor-level, model-level, and network-level attacks.

Sensor-Level Attacks

1. Poisoned vital-sign segments
2. Time-series injection (e.g., drift or spike insertion)
3. ECG/PPG sensor-spoofing attacks

Model-Level Attacks

4. Model poisoning (e.g., label flipping)
5. Backdoor pattern insertion
6. Free-rider clients submitting random activations

Federated Learning Attacks

7. Gradient-scaling adversaries
8. Sign-flipping adversaries
9. Byzantine update strategies

Network Attacks (TON_IoT)

10. Man-in-the-middle (MITM)
11. Distributed denial-of-service (DDoS)
12. Data exfiltration attacks

The quantum fidelity and trace-distance modules of HCQ-SFL-IR are expected to enhance detection performance, isolate malicious participants more rapidly, and reduce model degradation under attack.

Recommended Federated Partitioning Strategy

MIMIC-III / MIMIC-IV

- Client = individual ICU bed, or
- Client = hospital unit, or
- Client = sensor modality (ECG, PPG, BP)

WESAD

- Client = individual subject

TON_IoT

- Client = IoMT ward or device type

These partitions ensure strong non-IID distributions, which faithfully reflect real IoMT deployments and challenge the robustness of federated learning systems.

Summary

Purpose	Dataset
Primary Medical IoMT	MIMIC-III Waveform/Vitals
Wearable IoMT	WESAD
Activity IoMT	UCI HAR / PAMAP2
Network Intrusion	TON_IoT Healthcare Subset
Optional Additions	IoTID20, CIC-IDS2017

The attack suite includes model poisoning, gradient manipulation, sensor spoofing, backdoor triggers, adversarial temporal drift, and diverse network-level attacks (MITM, DDoS, and botnets). The proposed split-learning configuration employs lightweight CNN/TCN client encoders, PQC-based anomaly detectors on the server, and a quantum-weighted robust aggregation mechanism at the federated coordinator.

This comprehensive experimental plan ensures that HCQ-SFL-IR is evaluated across clinically realistic workloads and adversarial conditions, strengthening the novelty and rigor of the proposed research.